Finite-sample learning dynamics of discrimination recovery in amortized neural IRT

representation sharing degrades discrimination, decoupling restores it as a convergence-rate effect

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Amortized neural IRT

A sequence encoder E_ϕ (LSTM) maps a response history to a time-varying ability θ_t. Item parameters, discrimination α_i and difficulty or step thresholds β_i, are read from item embeddings and combined in an IRT decoder. For graded responses we use the GPCM (binary 2PL at $K{=}2$),

$$P(Y_i = k \mid \theta) \; = \; \frac{\exp \sum_{m=1}^{k} \alpha_i(\theta - \beta_{i,m})}{\sum_{l=0}^{K-1} \exp \sum_{m=1}^{l} \alpha_i(\theta - \beta_{i,m})}.$$

Training minimizes a response-prediction loss on the decoder (weighted ordinal cross-entropy). There is **no supervision** on θ, α, β and no parameter priors.

Recovery is the sign-aligned Spearman ρ between recovered and data-generating parameters. The latent scale is identified only up to an affine gauge ($\theta \to s\theta + t,\ \alpha \to \alpha/s$), so rank is the well-posed target, not magnitude.

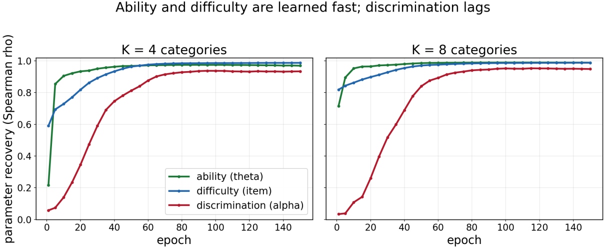
The architecture: shared vs decoupled item representation

The item code enters in two places, the **encoder input** (which forms θ) and the **discrimination/difficulty readouts**.

- **Shared** (`item_key_dim=None`, `emb_dim=`W). One item-embedding table of width W feeds both. Discrimination reads the same code that feeds ability.
- **Decoupled** (`emb_dim=8` + `item_key_dim=`$W-8$). A narrow item value embedding feeds the encoder, a *separate* wide item key embedding feeds only the α, β readouts.

In both arms `state_alpha=True`, so $\alpha_{i,t} = \text{head}([h_t, \text{key}_i])$ is state-conditioned and occurrence-averaged at recovery. The study varies `item_key_dim` (the representation) with `state_alpha` held on. The decoupled default also uses an exponential positivity map, shown later not to be the lever.

Discrimination is the low-information parameter



2PL gradients with residual $r = p - y$,

$$\partial_\theta L = r\alpha, \quad \partial_\beta L = -r\alpha, \quad \partial_\alpha L = r(\theta - \beta).$$

Fisher information, $w = p(1-p)$,

$$I(\theta) = I(\beta) = \alpha^2 w, \quad I(\alpha) = (\theta - \beta)^2 w.$$

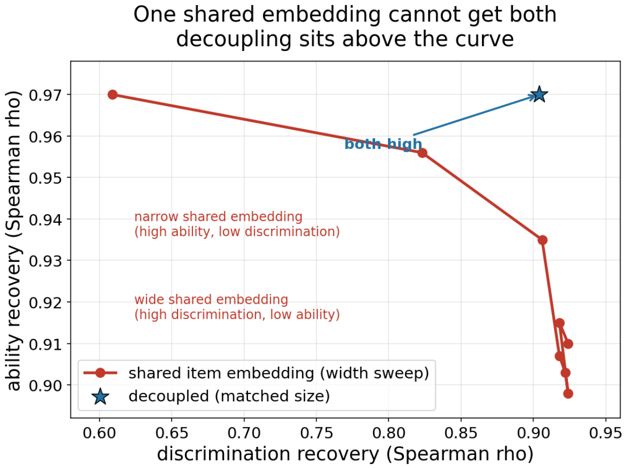
$I(\alpha)$ is suppressed by the lever arm and vanishes at $\theta = \beta$, where targeted responses concentrate. So α recovers slowly. The stiffness $\kappa = I(\theta)/I(\alpha)$ grows with K.

Three explanations for degraded discrimination under sharing

1. **Capacity.** The shared code is too small. A wider one fixes it. (Trivial.)
2. **Expressivity.** No shared code, however wide, can hold the θ-optimal and α-optimal directions at once. (A property of the global optimum, not of training.)
3. **Dynamics.** A shared code with the rank to express both is still driven by gradient flow to the θ compromise. (A finite-sample learning-dynamics effect.)

Each test below is designed to eliminate one explanation, with its control stated. Synthetic GPCM, known θ, α, β, LSTM encoder.

Test 1, allocation not budget (the gate)

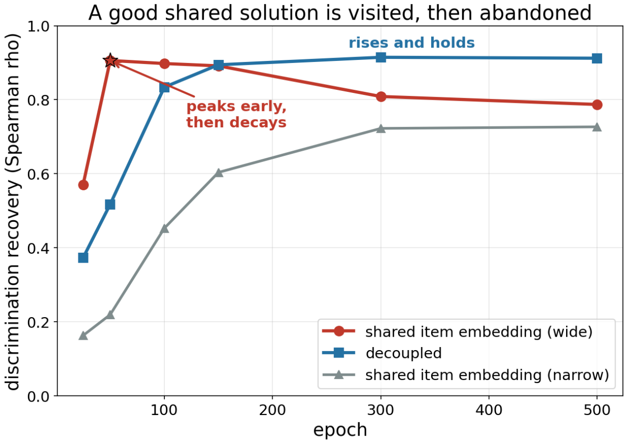


Design. Sweep the shared width W; place the decoupled model at **matched total item-embedding parameters** $(Q \times W)$ on the same axes. 5 seeds.

Result. The shared family traces a Pareto frontier, θ falls 0.97 $\rightarrow$ 0.88 as α climbs 0.66 $\rightarrow$ 0.91, and no width clears both. The decoupled point sits **above** the frontier, $\theta \approx 0.97$ and α 0.90 to 0.94, at fewer params.

Conclusion. Capacity eliminated. The effect is allocation, not budget.

Test 2, the both-high solution is reached then left

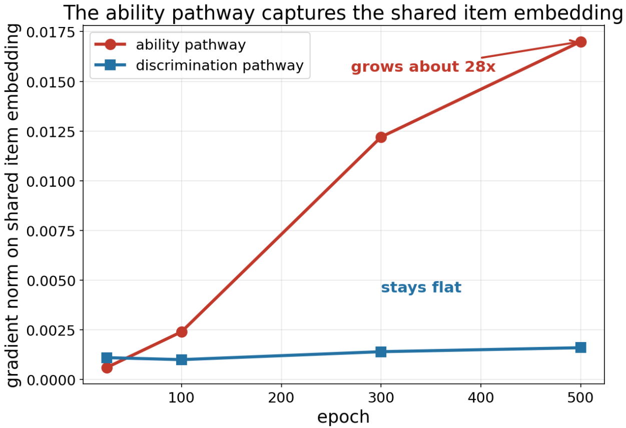


Design. Checkpoint training, recover α per epoch, for wide-shared, decoupled, narrow-shared. 3 seeds.

Result. Wide-shared α peaks at 0.906 (epoch 50, with θ also high) then **decays** to 0.787 by epoch 500. Decoupled rises to 0.912 and holds.

Conclusion. Expressivity eliminated. The both-high optimum is reachable and reached, then training leaves it. A dynamics effect, not a ceiling.

Test 2, mechanism, conditioning of the shared code

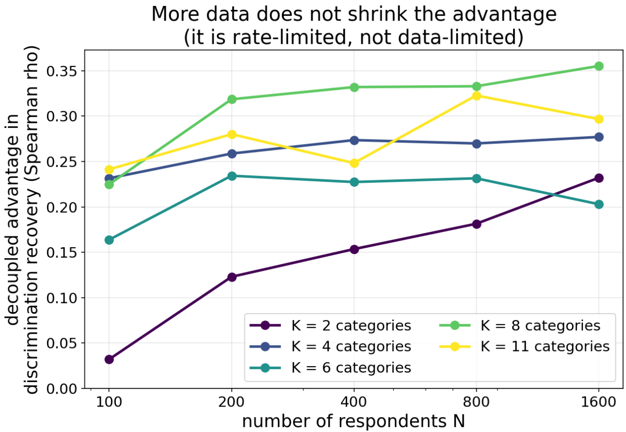


Design. Split the gradient on the shared item embedding e_q into the θ pathway and the α pathway, track over training (one instrumented run, sum-check exact).

Result. The θ-pathway gradient grows $\sim 28\times$ while the α-pathway stays flat, with $\cos(g_\theta, g_\alpha) \approx 0$.

Conclusion. Not a gradient tug of war. The mechanism is **Hessian conditioning**, the shared-code block's eigenvalue spread $\sim \kappa = I(\theta)/I(\alpha)$ (Hessian estimate ≈ 2.2 at $K=4$). The fast θ mode resolves first, the low-Fisher α mode last.

Test 3, the advantage is rate-limited, not data-limited

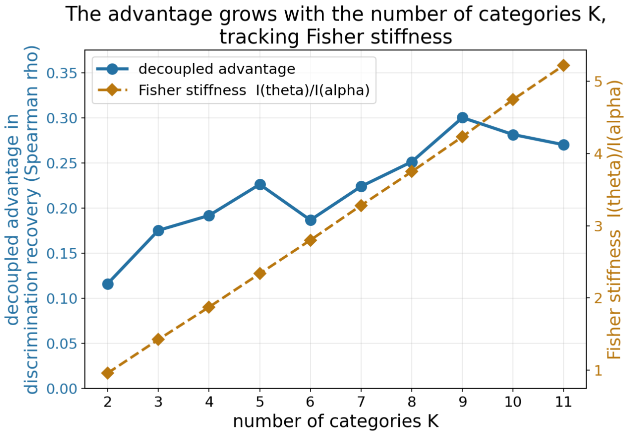


Design. Vary the number of learners N at a **fixed** 150-epoch budget, measure the decoupled advantage Δ_α, across K. 5 seeds.

Result. Δ_α does not shrink with N. It is flat to widening ($K{=}2$, $+0.03 \rightarrow +0.23$ over $N{=}100$ to 1600), higher for larger K.

Conclusion. The finite-sample signature. The effect is rate-limited, not data-limited, more data at a fixed budget leaves the stiff α mode further behind, decoupled keeps up. (At convergence it vanishes, see mechanism.)

Test 3, the advantage scales with Fisher stiffness

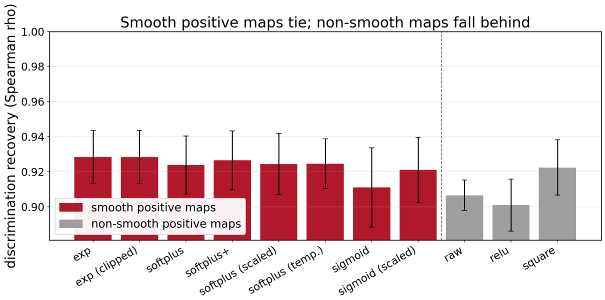


Design. Sweep K=2 to 11 (10 seeds), compare Δ_α to the analytic stiffness $\kappa = I(\theta)/I(\alpha)$.

Result. Δ_α tracks κ closely, $\rho(\Delta_\alpha, \kappa) = 0.89$ across K, and decoupled de-noises α (seed std 0.005 to 0.042). Rise then saturate, decoupled ceilings ≈ 0.95.

Conclusion. The size of the effect is governed by the Fisher stiffness, which the task (K) sets.

The positivity map is not the lever



α is read through a positive map $g(\cdot)$. Hypothesis, the exponential map is geometrically special.

Design. Matched effective-α initialization, per-map learning-rate sweep, $K = 2, 4, 8$. Plus a direct-α control with θ, β frozen, comparing α-space update rules.

Result. exp does not beat the best smooth map ($\Delta \approx 0$); only non-smooth or non-monotone maps (raw, ReLU, square) lag. The direct control refutes the α^2-preconditioner-only account.

Conclusion. Positivity is necessary, the exponential is not special. The lever is the representation, not the map.

What the theory pins

- Discrimination is the low-Fisher parameter, $I(\alpha) = (\theta - \beta)^2 w$ vanishes where responses concentrate, and the stiffness $\kappa = I(\theta)/I(\alpha)$ grows with K.
- Near an identifiable optimum, each direction's recovery rate is set by its Fisher information, so κ sets how late α resolves. **Low Fisher sets the rate.**
- **Free-table invariant.** When item parameters form a free table that can reach $p = p^*$, every gradient pull is linear in the residual and all vanish together, so the population optimum is unbiased and *decoupling-invariant*. **Low Fisher does not set the endpoint.**

So decoupling is a finite-budget convergence-rate and rank advantage, with no endpoint bias. Derivations in the supporting analysis.

Conclusion

Decoupling the discrimination representation in amortized neural IRT buys a Fisher-conditioning-governed convergence-rate advantage on discrimination rank recovery, growing with K, dominant at any practical training budget, with no endpoint cost.

- Not capacity (allocation at matched params), not expressivity (the both-high optimum is reached then left), but learning dynamics (the θ mode conditions the shared code, $\sim 28\times$).
- It is a measurement-validity condition, recovered discrimination is otherwise a representation-allocation artifact of finite-budget training.
- The positivity map is ruled out as the lever.

Scope. Synthetic GPCM, LSTM (architecture-independent on a swap bench), 3 to 10 seeds. **Open.** Per-parameter channel grid (θ, thresholds), NRM-slope channel, GRM head, real-data stability.